

Sai Amrith Reddy

Electronics and Computer Engineering Student

Specializing in AI/ML and Software Development. Built multiple scalable systems including ML pipelines, embedded systems, and full-stack applications. Focused on optimizing performance, automation, and real-world impact.



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amrithdev.vercel.app

EDUCATION

Bachelor of Technology (ECE) 2023 - 2027

Amrita Vishwa Vidyapeetham, Bengaluru | CGPA: 7.15

Registration No: BL.EN.U4EAC23051

Pre-University Course (PUC) 2022

Gopalan International School, Bangalore

Secondary Education (Class X) 2020

Sri Sathyasai Sharadanikethan

PROJECTS

AI Educational Video Generator

Python, LLM, Manim

Automated explainer video generation pipeline reducing manual effort by ~80%. Generated 2-minute videos with voiceovers, transitions, and mathematical visualizations using LLM scripting and Manim.

Deep Learning Music Generator

Python, PyTorch, Transformers, MIDI

Developed a transformer-based PyTorch model to generate multi-instrument MIDI tracks. Used attention mechanisms and relative positional encoding for tempo and structural consistency.

Student Mental Health Analysis

Python, Scikit-learn, MySQL

Built ensemble ML models (XGBoost, SVM, Random Forest) to improve predictive accuracy by 25%. Normalized and queried a MySQL backend database holding 10,000+ student profiles.

PhotoSpeakWrite

Python, Computer Vision, ML, Embedded Systems

Developed a hardware-software CV pipeline converting handwriting and speech to robotic arm instructions. Reduced serial communication latency by 40% using OpenCV.

Content Creation Campaigner

Python, Flask, LLMs, React, Presentation APIs

Designed an AI-powered tool generating platform-specific social media posts and slides from seed ideas and vibes. Built automated LLM prompt pipelines and templating workflows to produce polished multi-channel assets.

CCTV Streaming Platform

Python, Flask, REST APIs

Built Flask REST APIs for low-latency CCTV video streaming. Implemented JWT token authentication, role-based access control, and multi-threaded buffering.

TECHNICAL ACTIVITIES

Smart India Hackathon (SIH) 2024

Developed an SVM-based classification system for real-time target identification to distinguish between drones and birds. Trained the model and deployed it on embedded hardware for low-latency detection in dynamic environments.

GDG DevFest Workshop Facilitator 2024 - Present

Conducted hands-on technical workshops on React, Firebase, and Kotlin for 100+ students. Designed open-source coding exercises and demo templates.

National Cyber Hackathon Participant 2023

Competed in cybersecurity vulnerability assessment challenges. Solved complex reverse engineering, cryptography, and web exploitation problems in a team format.

SKILLS

Python

Java

C

Kotlin

PyTorch

TensorFlow

Scikit-learn

Flask

React

REST APIs

Git

Linux

MySQL

Manim

Problem Solving

Team Work

Communication

Adaptability

Critical Thinking

CERTIFICATIONS

UI/UX Design Certification

UI principles, wireframing, prototyping, and user experience best practices.

Java Developer Certification

Core Java development, collections, exception handling, multithreading, and architecture.

Java OOP Certification

Object-oriented programming: inheritance, polymorphism, encapsulation, and design patterns.

Machine Learning Specialization

Supervised and unsupervised learning, decision trees, and neural networks.

LEADERSHIP POSITIONS

Core Team - GDG On Campus

2024 - Present

Google Developer Group, Amrita Bengaluru

Web & App Dev Team Lead. Organized workshops, hackathons, and bootcamps on modern web, mobile, and cloud tech. Mentored 50+ peers in building production-ready apps.

LANGUAGES

English

- Full Professional Proficiency

Telugu

- Native or Bilingual Proficiency

Hindi

- Limited Working Proficiency

AREAS OF INTERESTS

AI / ML

Computer Vision

Embedded Systems

NLP

Full-Stack Dev

Automation